Floquet Yao-Kivelson

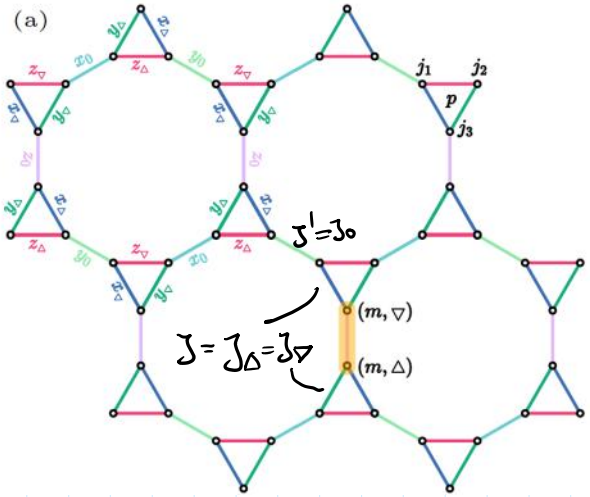
Mittwoch, 15. Juli 2026 16:30

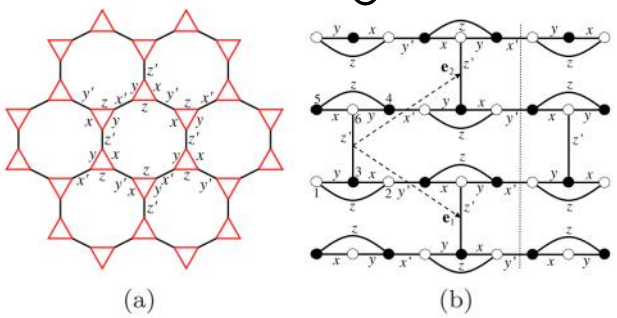
goal: generalize the Yao-Kivelson model to Floquet

Hamiltonian version: exactly solvable, chiral spin liquid, non-abelian anyons (Majorana bound states)

hope for Floquet: non-abelian Floquet anyon transmutation

very similar to IBM's device!





Hamiltonian: Yao, Kivelson '08; Bauer, Reuther '26

$$H = J_\Delta \sum'_{\langle ij\rangle_\Delta} \sigma_i^\alpha \sigma_j^\alpha + J_\nabla \sum'_{\langle ij\rangle_\nabla} \sigma_i^\alpha \sigma_j^\alpha + J_0 \sum'_{\langle i,j\rangle_0} \sigma_i^\alpha \sigma_j^\alpha$$

$J = J_\Delta = J_\nabla$, $J' = J_0$ $\quad \sigma_i^\alpha = i b_i^\alpha c_i$

gauge constraint: $\sigma^x\sigma^y = i\sigma^z$

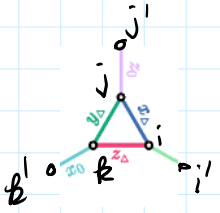
$$1 = -i\sigma^x\sigma^y\sigma^z = -i(ib^xc)(ib^yc)(ib^zc) =$$

$$= i^4 b^x b^y c^2 b^z c$$

$$= b^x b^y b^z c$$

$$\sigma_i^\alpha \sigma_j^\alpha = (ib_i^\alpha c_i)(ib_j^\alpha c_j) = -(ib_i^\alpha b_j^\alpha)(ic_i c_j)$$
$$= -u_{ij}^{\alpha\alpha}(ic_i c_j)$$

fluxes: • $W_\Delta = \sigma_i^x \sigma_j^x \sigma_j^y \sigma_k^y \sigma_k^z \sigma_i^z$



$[H, W_\Delta] = 0$

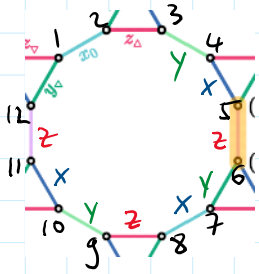
since $\sigma_j^z \sigma_{j'}^z$, $\sigma_i^y \sigma_{i'}^y$, $\sigma_k^x \sigma_{k'}^x$ commute

analogous for W_∇

$$W_\Delta^2 = (\sigma_i^x \sigma_j^x \sigma_j^y \sigma_k^y \sigma_k^z \sigma_i^z)(\sigma_i^x \sigma_j^x \sigma_j^y \sigma_k^y \sigma_k^z \sigma_i^z)$$
$$\overset{\leftarrow -1 \rightarrow}{=} (-1)^2 (\sigma_j^y \sigma_k^y \sigma_k^z \sigma_i^z)(\sigma_j^y \sigma_k^y \sigma_k^z \sigma_i^z)$$
$$= -(\sigma_k^z \sigma_i^z)^2$$
$$= -1$$

$$\boxed{W_{\Delta,\nabla} = \pm i} \quad !$$

odd number of bonds



$$W_0 = (\sigma_1^x \sigma_2^x)(\sigma_2^z \sigma_3^z)(\sigma_3^y \sigma_4^y)\ldots$$

conserved $[H, W_0] = 0$

$W_0^2 = 1$ $\quad \boxed{W_0 = \pm 1}$

$$W_\Delta = (-i)^3 \prod_{\langle ij \rangle_\Delta} u_{ij}^{\alpha\alpha} \quad \begin{cases} i & -\pi/2 \text{ flux} \\ -i & \pi/2 \text{ flux} \end{cases}$$

$$W_0 = (-i)^{12} \prod_{\langle ij \rangle_0} u_{ij}^{\alpha\alpha} = \begin{cases} 1 & \pi \text{ flux} \\ -1 & 0 \text{ flux} \end{cases}$$

mapping yields:

$$H = i \sum_{jk} t_{jk} c_j c_k$$

$$t_{jk} = - u_{jk} J_\chi \qquad \chi = 0, \Delta/\nabla$$

sign of hopping corresponds to direction of gauge link

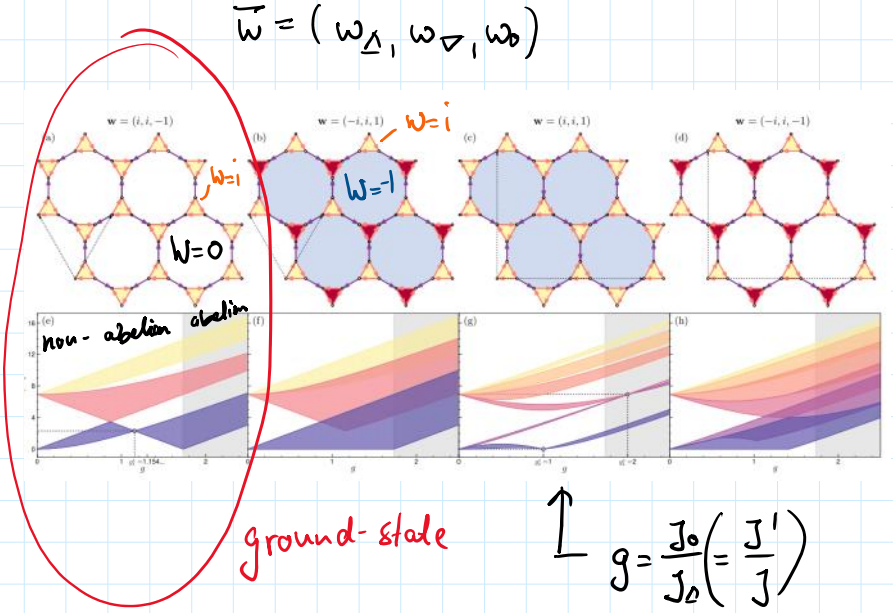
groundstate?

Lieb's theorem: not applicable (bipartite)

choose uniform flux configuration:

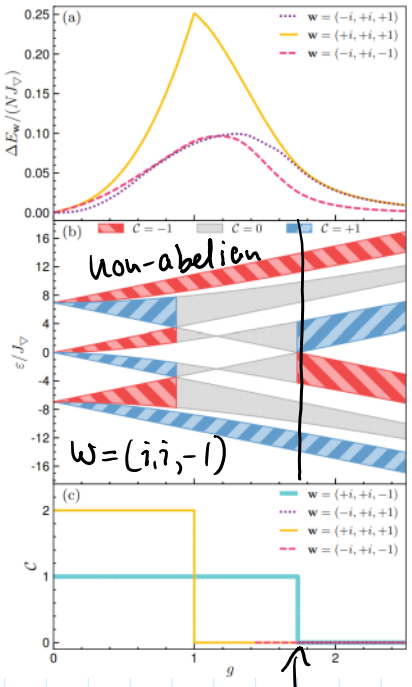
$$\omega_\Delta = \pm i, \quad \omega_\nabla = \pm i, \quad \omega_\circ = \pm 1$$

$$\vec{w} = (\omega_\Delta, \omega_\nabla, \omega_\circ)$$



ground-state

$g = \frac{J_\circ}{J_\Delta} \left(= \frac{J'}{J} \right)$



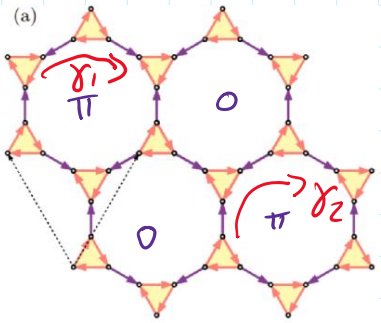
abelian

$C_{tot} = 0$

$C_{tot} = 1$

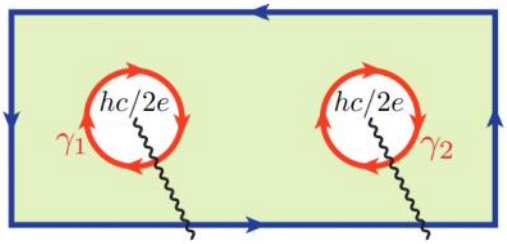
$g_c = \sqrt{3}$

Non-abelian anyons : $(J \gg J')$



$\hat{=}$

p+ip Superconductor, Alicea, 2013



Vao, Hirsch:

we have confirmed that the ground state degeneracy for 2n vortices on a torus is 2n+1, independent of their positions or whether the vortices are associated with a triangular or a dodecagonal plaquette, so long as the vortices are separated by a distance large compared to the inverse energy gap.

It is to be expected that the vortex excitations in a state with odd Chern number obey non-Abelian braiding statistics [10], similar to the physics of p + ip superconductors [25] and the Pfaffian state of 5/2-FQHE [26]. Specifically, in the odd Chern number phase, each vortex carries a Majorana zero energy mode. A pair of Majorana zero-energy modes on two well separated vortices constitute a single complex fermionic mode with zero energy.

In the small J (v = 0) phase, the vortices have Abelian braiding statistics. When $J \ll J'$, by mapping the model to an effective Z2 gauge theory on a "dual" honeycomb lattice with spontaneously broken TRS [21], we find the vortex anyons on both triangle and dodecagon plaquettes are bosons. Interestingly, they are relative semions

FLOQUET : let's try:

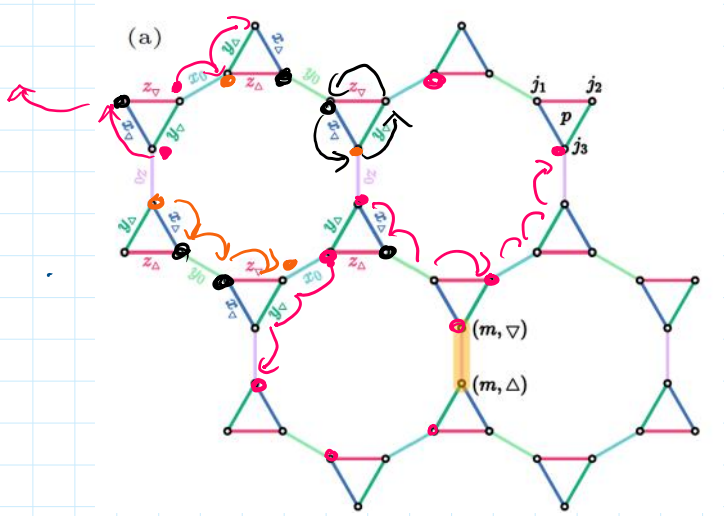
$$U_F = \left(e^{iXX_\Delta} e^{iXX_\nabla} e^{iXX_O}\right) \quad \text{all } XX$$

$$\times \left(e^{iW_\Delta} e^{iW_\nabla} e^{iW_O}\right) \quad - YY$$

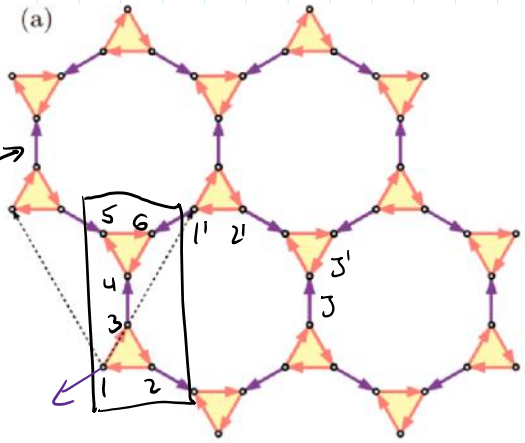
$$\times \left(e^{iZZ_\Delta} e^{iZZ_\nabla} e^{iZZ_O}\right) \quad - ZZ$$

$\chi(e^{i2\pi\Delta} e^{i2\pi\nabla} e^{i2\pi\square}) = 2\pi$

- encircel other plaquettes
- encircle O in 4 steps
- localized on Δ / ∇



implement Bloch Hamiltonian:



arrows: direction of gauge links

$u_{jk} \hat{=} \longrightarrow = +1$

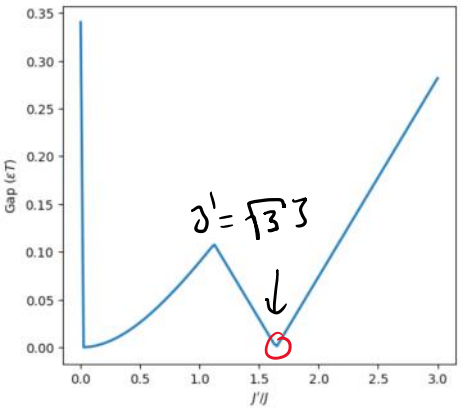
$= -u_{kj}$

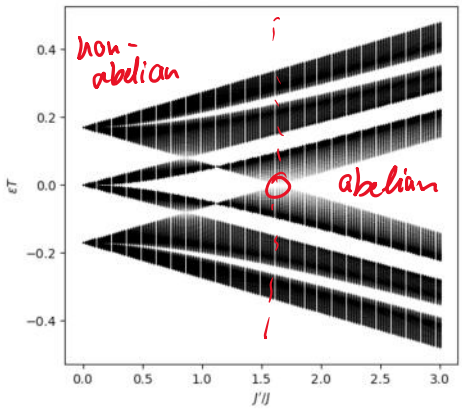
within unit cell: $2\to1$, $1\to3$, $3\to2$, $3\to4$, $4\to5$, $5\to6$, $6\to4$

other unit cells: $6\leftarrow 1'$, $5\leftarrow 2''$, $1\to 6'''$, $2\to 5''''$

$+\vec{a}_1$, $\vec{a}_2$, $-\vec{a}_1$, $-\vec{a}_2$

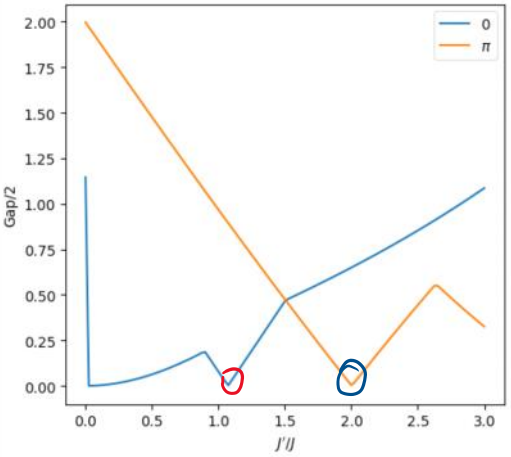
Bandstructure: $T=0.1$, high-frequency limit $\sim$ Hamiltonian

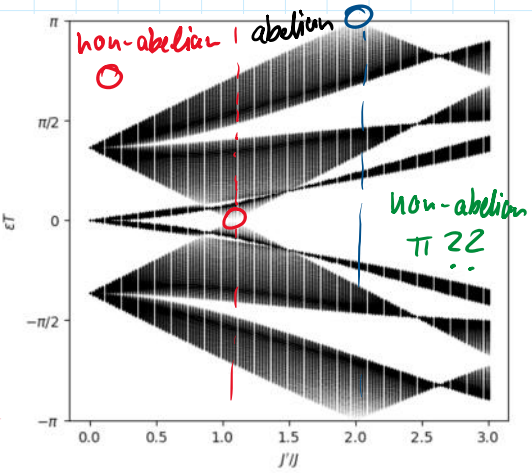




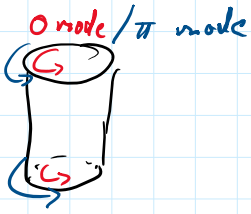
check ✓

T=0.8 :

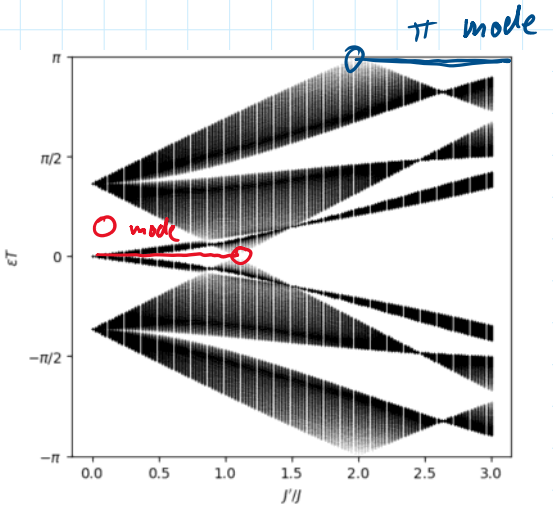




phases should follow from open boundary conditions



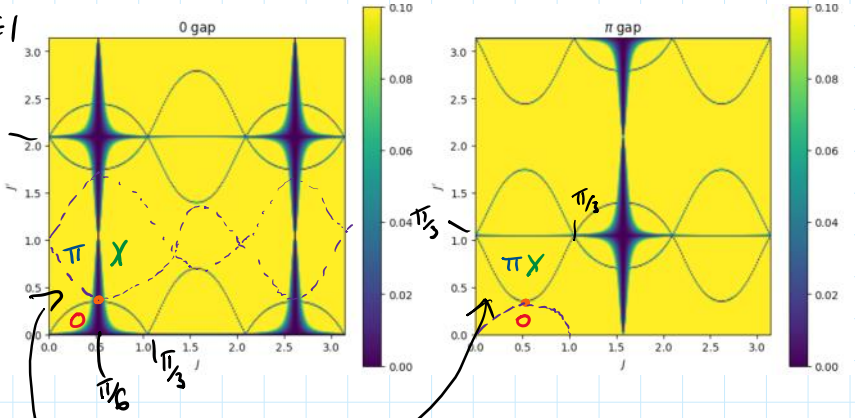
I suspect this should happen for OBC:



gap closings: I see numerically that they happen always at the Γ point:

gap closings: I see numerically that they happen always at the Γ point:

T=1



trivial

X sweet spot, $J_1 = J_2 = \pi/4$, where

$U_F = e^{i\frac{\pi}{4}XX} e^{i\frac{\pi}{4}YY} e^{i\frac{\pi}{4}ZZ}$!